

Probability Review

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Definitions

Imagine you have a large number N of similarly prepared systems, an *ensemble*, and you conduct a measurement. If there is insufficient information to determine the outcome, you may still be able to state the probability of the measurement. The probability of the result being between x and $x + dx$ is

$$\begin{aligned} p(x)dx &= \lim_{N \rightarrow \infty} \frac{\# \text{ of results between } x \text{ and } x + dx}{N} \\ &= \text{prob}(\text{single system result between } x \text{ and } x + dx) \end{aligned}$$

The “random variable” is x .

Properties

Continuous

$$0 \leq p(x)$$

$$p(a < x < b) = \int_a^b p(x) dx$$

$$p(\text{anywhere}) = \int_{-\infty}^{\infty} p(x) dx = 1$$

probability density $p(x)$ can be greater than 1

Cumulative probability density $P(x) = \int_{-\infty}^x p(y) dy$

$$p(x) = \frac{d}{dx} P(x)$$

averages of x or functions of x can be calculated $\langle F(x) \rangle = \int_{-\infty}^{\infty} F(x) p(x) dx$

$$\text{mean value} = \langle x \rangle = \int_{-\infty}^{\infty} x p(x) dx$$

$$n^{\text{th}} \text{ moment } \langle x^n \rangle = \int_{-\infty}^{\infty} x^n p(x) dx$$

$$\text{Variance } \text{var}(x) = \langle x^2 \rangle - \langle x \rangle^2 = \langle (x - \langle x \rangle)^2 \rangle$$

$$\text{Standard deviation } \sigma = \sqrt{\text{var}(x)}$$

Discrete

$$0 \leq p_j \leq 1$$

$$p(a \leq j \leq b) = \sum_{j=a}^b p_j$$

$$1 = \sum_{\text{all } j} p_j$$

$$\langle F(j) \rangle = \sum_j F(j) p_j$$

$$\langle j \rangle = \sum_j j p_j$$

$$\langle j^n \rangle = \sum_j j^n p_j$$

Normal distribution = Gaussian $p(x)$

Data taken with mean μ and standard deviation σ often approaches the Gaussian distribution:

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

The cumulative probability distribution is related to the *error function*

$$P(x) = \int_{-\infty}^x p(y) dy = \frac{1}{2} \left(1 + \text{erf}\left(\frac{x-\mu}{\sqrt{2}\sigma}\right) \right)$$

Binomial distribution

Suppose there are two possible outcomes to a measurement: A and $\bar{A}$, such as heads or tails of a (possibly weighted) coin. Define the probability of measuring A to be p and the probability of measuring $\bar{A}$ to be $q = 1 - p$. If you repeat the experiment N times, what is the probability of A occurring n times, i.e. $p_N(n)$? If one is flipping a fair coin (so $p = 1/2$) then: For $N = 1$, one has $p_1(0) = p_1(1) = \frac{1}{2}$ and for $N = 2$ one has $p_2(0) = p_2(2) = \frac{1}{4}$ and $p_2(1) = \frac{1}{2}$. For general N , what should one do? First notice that each measurement results in either a p or a q .

$$1 = (p + q)^N = \sum_{n=0}^N \frac{N!}{n! (N-n)!} p^n q^{N-n} \quad (1)$$

So the probability of n events A out of N trials is

$$p_N(n) = \frac{N!}{n! (N-n)!} p^n q^{N-n}$$

To find the average number of occurrences of A

$$\langle n \rangle = \sum n p_N(n) = \sum n \frac{N!}{n! (N-n)!} p^n q^{N-n} \quad (2)$$

Here's a useful trick. Note that if one differentiates the RHS of (1) by p , one brings down a factor of n while reducing the power of p by one. So if one then multiply the result by p one gets (2). In symbols $p \frac{\partial}{\partial p} \leftrightarrow n$.

Let's see how it works.

$$\begin{aligned} \langle n \rangle &= p \frac{\partial}{\partial p} \sum_{n=0}^N \frac{N!}{n! (N-n)!} p^n q^{N-n} \\ &= p \frac{\partial}{\partial p} (p + q)^N = p N (p + q)^{N-1} = p N \end{aligned} \quad (3)$$

After differentiating, we set $(p + q) = 1$.

With this you can calculate the second moment

$$\langle n^2 \rangle = \left(p \frac{\partial}{\partial p} \right) \left(p \frac{\partial}{\partial p} \right) (p + q)^N = Np + N(N-1)p^2$$

You will show this in your homework and also show that the standard deviation is $\sqrt{Npq}$.

Poisson distribution

Another important distribution, relevant for “rate processes” like radioactivity, describes events occurring randomly in time. The distribution is uniquely determined by the rate r at which events occur. Consider an interval of time dt , chosen small enough so that the probability of more than one event during dt is negligible. The probability of an event during this dt is $p(1; dt) = r dt$ and the probability of no event during this dt is $p(0; dt) = 1 - r dt$. What is the probability that n events will occur during a long time interval t , aka $P(n; t)$?

To figure this out, we will start by trying to calculate $P(n; t + dt)$ and recognizing that this can be split apart into two time intervals: from 0 to t and from t to $t + dt$. At most, a single event is in the second interval, with all others necessarily in the first. Mathematically,

$$\begin{aligned} P(n; t + dt) &= P(n - 1; t)P(1; dt) + P(n; t)P(0; dt) \\ &= P(n - 1; t)r dt + P(n; t)(1 - r dt) \end{aligned} \quad (4)$$

Taking the difference, we have

$$P(n; t + dt) - P(n; t) = r dt (P(n - 1; t) - P(n; t))$$

so the derivative is clearly

$$\frac{dP(n; t)}{dt} = r (P(n - 1; t) - P(n; t)) \quad (5)$$

To solve this set of equations, we need initial conditions. For any rate process, if no time has elapsed, no event have been detected. This means

$$\begin{aligned} P(0; 0) &= 1 \\ P(n; 0) &= 0 \quad \text{for } n > 0 \end{aligned}$$

In particular, for $n = 0$,

$$\frac{dP(0; t)}{dt} = -rP(0; t)$$

is solved by

$$P(0; t) = \exp(-rt)$$

For general n , you will show that $P(n; t) = \frac{(rt)^n}{n!} \exp(-rt)$ solves the differential equation (5). You will also calculate $\langle n \rangle$ and the standard deviation of n .